

List of Publications

Bastien Carreres

Publications

Publications as first author

1. **Carreres et al.** 2025. *Type Ia Supernova Growth-rate Measurement with LSST Simulations: Intrinsic Scatter Systematics*
Published in The Astrophysical Journal. doi: 10.3847/1538-4357/ae11ac
In this paper, I study the impact of intrinsic scatter of SNe Ia on the measurement of $f\sigma_8$. This study is done through the simulation of the full low- z LSST SNe Ia sample for different intrinsic scatter models. We found that the most realistic model of intrinsic scatter causes non-Gaussianities in the Hubble diagram residuals, resulting in a bias on $f\sigma_8$.
2. **Carreres et al.** 2025. *ZTF SN Ia DR2: Peculiar velocities' impact on the Hubble diagram*
Published in Astronomy & Astrophysics. doi: 10.1051/0004-6361/202450389
This paper is part of the second data release of ZTF SNe Ia. In this paper, I study the impact of the peculiar velocity (PV) systematics on the Hubble diagram of the ZTF SN Ia DR2 sample. We show that not taking into account the full PV covariance matrix can lead to a slight underestimation of the error on the Hubble constant H_0 and could shift its value by $\sim 1 \text{ km s}^{-1}$.
3. **Carreres et al.** 2023. *Growth-rate measurement with type-Ia supernovae using ZTF survey simulations*
Published in Astronomy & Astrophysics. doi: 10.1051/0004-6361/202346173
This paper is the main publication of my thesis. In this paper, I present my work to prepare for the future measurement of the growth rate of structure ($f\sigma_8$) from ZTF SN Ia data. I describe my realistic simulation of the ZTF SN Ia sample and, using these simulations, I study the biases and systematics that can affect the measurement of $f\sigma_8$. I show that using SN Ia data from the full 6 years of the ZTF II sample with a cut at a redshift of $z < 0.06$ to avoid selection due to magnitude limit, we can expect an unbiased measurement of $f\sigma_8$ with an error of $\sim 19\%$.

Publications with significant contribution

1. Acevedo, Scolnic, **Carreres et al.** 2026. *WHALES. The Weighing Halos Accurately, Locally, and Efficiently with Supernovae Survey: Overview and Initial Data Release*
Published in The Astrophysical Journal. doi: 10.3847/2041-8213/ae3d0b
This paper introduces the WHALES survey that is discovering SNe Ia nearby the Shapley Supercluster. I contributed to simulations and forecasts described in this paper.
2. Acevedo, Sherman, Brout, **Carreres et al.** 2025. *The Dark Energy Bedrock All-sky Supernova Program: Cross Calibration, Simulations, and Cosmology Forecasts*
Published in The Astrophysical Journal. doi: 10.3847/1538-4357/ae1e78
This publication introduces the Dark Energy Bedrock All-Sky Supernova (DEBASS) low- z survey. The paper gives an overview of the survey, examines systematic effects such as photometric calibration and selection effects, presents simulations, and provides forecasts for future cosmological analysis. I wrote and ran the forecast codes, and I was actively involved in the simulations and writing process.
3. Acevedo, Scolnic, **Carreres et al.** 2025. *The Weighing Halos Accurately, Locally, and Efficiently with Supernovae (WHALES) Survey Overview and Initial Data Release*
Pre-published on arXiv. doi: 10.48550/arXiv.2510.26731
This paper introduces the WHALES survey that is discovering SNe Ia nearby the Shapley Supercluster. I contributed to simulations and forecasts described in this paper.

4. Rosselli, **Carreres** et al. 2025. *Forecast for growth-rate measurement using peculiar velocities from LSST supernovae*
Published in Astronomy & Astrophysics. doi: 10.1051/0004-6361/202556181
This paper presents a simulation-based forecast for the measurement of the growth rate of structure ($f\sigma_8$) with the Rubin-LSST survey. It shows that the full constraining power of the Rubin-LSST SNe Ia sample will enable a constraint at the $\sim 10\%$ level in the redshift range $0.02 < z < 0.14$. I participated in the development of the simulation and analysis code used in this publication. I was also involved in the writing and review process.
5. Ravoux, **Carreres** et al. 2025. *Generalized framework for likelihood-based field-level inference of growth rate from velocity and density fields*
Published in Astronomy & Astrophysics. doi: 10.1051/0004-6361/202554319
This paper presents the FLIP Python library. This library is based on codes developed during my PhD and proposes a more general framework for the constraint of the growth rate of structure. I actively participated in the development of FLIP and in the writing of this paper.
6. Peterson, **Carreres** et al. 2025. *Improving the Determination of Supernova Cosmological Redshifts by Using Galaxy Groups*
Published in The Astrophysical Journal. doi: 10.3847/1538-4357/ada285
In this paper, we used SNe Ia data and the Uchuu UniverseMachine simulation to study the improvement on the SNe Ia Hubble diagram that we can expect from averaging redshift over galaxy groups of SN Ia hosts. I ran the simulations and was actively involved in the analysis.

Publications as co-author

1. Anumba, Jones, Kessler, Scolnic, Kenworthy, Chen, **Carreres** et al. 2026. *Calibration-Induced Systematics in SALT3 Training and Their Impact on Dark Energy Constraints from Stage IV Supernova Surveys*
Pre-published on arXiv e-prints. doi: 10.48550/arXiv.2604.19746
I contributed to this paper by giving advice and comments on the analysis.
2. Barjou-Delayre, Rosnet, Ravoux, Aubert, Ginolin, Kebadian, Amenouche, Bautista, Burgaz, **Carreres** et al. 2026. *Study of the anisotropy of cosmic expansion on ZTF type Ia supernovae simulations*
Pre-published on arXiv. doi: 10.48550/arXiv.2601.11139
ZTF SN Ia cosmology group collaboration paper. My simulations of the ZTF survey are used in this paper.
3. Dixon, Jones, Drakos, **Carreres** et al. 2026. *The Peculiar Growth of Structure: Validating $f\sigma_8$ Measurements from TITAN Type Ia Supernovae and the Uchuu Simulations*
Pre-published on arXiv e-prints. doi: 10.48550/arXiv.2606.14962
I contributed to this paper by setting up the simulation and giving advice and comments on the analysis.
4. Popovic, Grayling, O’Callaghan, Smith, Boyd, Mandel, Wiseman, **Carreres** et al. 2026. *A Simulation Based Inference Approach to Modelling of Type Ia Supernova Populations*
Pre-published on arXiv e-prints. doi: 10.48550/arXiv.2607.28725
This paper presents a simulation-based inference (SBI) model of the dust affecting SN Ia. It was developed after I identified several issues in the method previously used to derive dust parameters in the DES analyses. I provided comments and feedback on this analysis.
5. Lacroix, Regnault, Jaeger, Jeune, Betoule, Colley, Bernard, Rigault, Smith, Goobar, Maguire, Dimitriadis, Nordin, Johansson, Aubert, Barjou, Bellm, Bongard, Burgaz, **Carreres** et al. 2025. *ZTF SNe Ia DR2: Towards cosmology-grade ZTF supernova light curves using scene modeling photometry*
Pre-published on arXiv e-prints. doi: 10.48550/arXiv.2509.04073
ZTF SN Ia cosmology group collaboration paper.
6. Ginolin, Rigault, Smith, Copin, Ruppin, Dimitriadis, Goobar, Johansson, Maguire, Nordin, Amenouche, Aubert, Barjou-Delayre, Betoule, Burgaz, **Carreres** et al. 2025. *ZTF SN Ia DR2: Environmental dependencies of stretch and luminosity for a volume-limited sample of 1000 type Ia supernovae*
Published in Astronomy and Astrophysics. doi: 10.1051/0004-6361/202450378
ZTF SN Ia cosmology group collaboration paper.

7. Amenouche, Rosnet, Smith, Rigault, Aubert, Barjou-Delayre, Burgaz, **Carreres** et al. 2025. *ZTF SN Ia DR2: Simulations and volume-limited sample*
Published in Astronomy and Astrophysics. doi: 10.1051/0004-6361/202452134
ZTF SN Ia cosmology group collaboration paper.
8. Aubert, Rosnet, Popovic, Ruppini, Smith, Rigault, Dimitriadis, Goobar, Johansson, Barjou-Delayre, Burgaz, **Carreres** et al. 2025. *ZTF SN Ia DR2: Exploring SN Ia properties in the vicinity of under-dense environments*
Published in Astronomy and Astrophysics. doi: 10.1051/0004-6361/202450951
ZTF SN Ia cosmology group collaboration paper.
9. Ginolin, Rigault, Copin, Popovic, Dimitriadis, Goobar, Johansson, Maguire, Nordin, Smith, Aubert, Barjou-Delayre, Burgaz, **Carreres** et al. 2025. *ZTF SN Ia DR2: Colour standardisation of type Ia supernovae and its dependence on the environment*
Published in Astronomy and Astrophysics. doi: 10.1051/0004-6361/202450943
ZTF SN Ia cosmology group collaboration paper.
10. Popovic, Rigault, Smith, Ginolin, Goobar, Kenworthy, Ganot, Ruppini, Dimitriadis, Johansson, Amenouche, Aubert, Barjou-Delayre, Burgaz, **Carreres** et al. 2025. *ZTF SN Ia DR2: Evidence of changing dust distribution with redshift using type Ia supernovae*
Published in Astronomy and Astrophysics. doi: 10.1051/0004-6361/202450391
ZTF SN Ia cosmology group collaboration paper.
11. Rigault, Smith, Goobar, Maguire, Dimitriadis, Johansson, Nordin, Burgaz, Dhawan, Sollerman, Regnault, Kowalski, Nugent, Andreoni, Amenouche, Aubert, Barjou-Delayre, Bautista, Bellm, Betoule, Bloom, **Carreres** et al. 2025. *ZTF SN Ia DR2: Overview*
Published in Astronomy and Astrophysics. doi: 10.1051/0004-6361/202450388
ZTF SN Ia cosmology group collaboration paper.
12. Rigault, Smith, Regnault, Kenworthy, Maguire, Goobar, Dimitriadis, Johansson, Amenouche, Aubert, Barjou-Delayre, Bellm, Burgaz, **Carreres** et al. 2025. *ZTF SN Ia DR2: Study of Type Ia supernova light-curve fits*
Published in Astronomy and Astrophysics. doi: 10.1051/0004-6361/202450377
ZTF SN Ia cosmology group collaboration paper.
13. Ruppini, Rigault, Ginolin, Dimitriadis, Goobar, Johansson, Maguire, Nordin, Smith, Aubert, Biedermann, Copin, Burgaz, **Carreres** et al. 2025. *ZTF SN Ia DR2: Impact of the galaxy cluster environment on the stretch distribution of Type Ia supernovae*
Published in Astronomy and Astrophysics. doi: 10.1051/0004-6361/202450956
ZTF SN Ia cosmology group collaboration paper.
14. Scolnic, Riess, Murakami, Peterson, Brout, Acevedo, **Carreres** et al. 2025. *The Hubble Tension in Our Own Backyard: DESI and the Nearness of the Coma Cluster*
Published in The Astrophysical Journal. doi: 10.3847/2041-8213/ada0bd
I contributed to this paper by running a check on an N-body simulation to determine the typical size of a Coma-like cluster.

Talks

1. Jun. 2026 – **Red Dust Redemption: A bayesian model of dust and stretch with the DEBASS and DES samples**, at *Oxford Next Generation of SN Survey workshop 2026*, University of Oxford, Oxford, UK
2. Apr. 2026 – **Cosmology with SNe Ia: measuring growth-rate of structures with the new generation of survey** at *Yale cosmology group seminar*, Yale University, New Haven, CT, USA
3. Jan. 2026 – **SNe Ia growth-rate measurements with Rubin-LSST simulations: intrinsic scatter systematics** at *CPPM Renoir science meeting*, Marseille, France

4. Nov. 2025 – **DESC Project announcement: Constraining Structure Growth and Modified Gravity Using LSST low-z SN Ia PVs and Weak Lensing** at *DESC-MCP meeting*, online
5. Sept. 2025 – **Status of PV cosmology in LSST-DESC** at *DESC workshop on cosmology with peculiar velocities*, Marseille, France
6. Jul. 2025 – **SNe Ia growth-rate measurements with Rubin-LSST simulations: intrinsic scatter systematics** at *Cosmic Lighthouses 2025*, Cambridge, UK
7. Feb. 2025 – **Improving SN Ia Hubble residual scatter with galaxy groups** at *CosmicFlows 2025*, Brisbane, Australia
8. Mar. 2025 – **SNe Ia growth-rate measurements with Rubin-LSST simulations : intrinsic scatter systematics** at *SN Ia cosmology group of LPCA*, online
9. Feb. 2025 – **SNe Ia growth-rate measurements with Rubin-LSST simulations : intrinsic scatter systematics** at *CosmicFlows 2025*, Brisbane, Australia
10. Oct. 2024 – **DESC Project announcement: Measurement of the growth-rate of structures using SN Ia PVs in the BBC framework** at *DESC-TD biweekly meeting*, online
11. Sept. 2023 – **Possible velocity systematic on the Hubble diagram fit** at *ZTF France*, LPC, Clermont-Ferrand, France
12. Aug. 2023 – **$f\sigma_8$ measurement with type Ia supernovae** at *DESC-TD biweekly meeting*, online
13. May 2023 – **Growth-rate measurement with type Ia supernovae** at the *Duke cosmology group' seminar*
14. Nov. 2022 – **Cosmology with the growth rate using type Ia supernovae** at *Action Dark Energy 2022*, Marseille, France
15. May 2022 – **Measuring $f\sigma_8$ with the ZTF SN Ia sample** at *Rubin-LSST France 2022*, LAPP, Annecy, France
16. May 2022 – **Measuring $f\sigma_8$ with the ZTF SN Ia sample** at *ZTF spring meeting*, LPNHE, Paris, France
17. June 2021 – **Peculiar velocities with Type Ia Supernovae** at *Rubin-LSST France 2021*, LPSC, Grenoble, France

Posters

1. 2022 - **Cosmology with the growth rate of structures using type Ia supernovae** at *DESC Summer Meeting 2022*, University of Chicago
2. 2022 - **Cosmology with the growth rate of structures using type Ia supernovae + Proceedings** at *Rencontres de Moriond*